Getting started with algebraicodin

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2026-03-23

Introduction

algebraicodin uses applied category theory to build epidemiological models compositionally, then generates odin2 code for high-performance simulation via dust2. Instead of writing ODE equations by hand, you define a *Petri net*—a bipartite graph of species (compartments) and transitions (processes)—and the package derives the equations automatically.

This vignette walks through a basic SIR (Susceptible–Infectious–Recovered) model, comparing the algebraic approach with hand-written odin2 code.

```
library(algebraicodin)
#>
#> Attaching package: 'algebraicodin'
#> The following objects are masked from 'package:base':
#>
#>     %o%, %x%
```

Defining a Petri net

A **labelled Petri net** describes a reaction network with named species and named transitions. Each transition has input arcs (consumed species) and output arcs (produced species).

For an SIR model, we have:

- **Species**: S (susceptible), I (infectious), R (recovered)
- **Transitions**:
  - `inf`: $S + I \to I + I$ (infection: one S and one I produce two I)
  - `rec`: $I \to R$ (recovery)

```
sir <- labelled_petri_net(
  c("S", "I", "R"),
  inf = c("S", "I") %=>% c("I", "I"),
  rec = "I" %=>% "R"
)
```

We can inspect the structure:

```
species_names(sir)
#> [1] "S" "I" "R"
transition_names(sir)
#> [1] "inf" "rec"
transition_matrices(sir)
#> $input
#>   inf rec
#> S   1   0
#> I   1   1
```

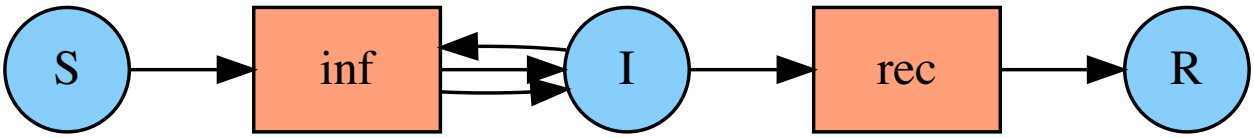
```
#> R   0   0
#>
#> $output
#>   inf rec
#> S   0   0
#> I   2   0
#> R   0   1
#>
#> $stoichiometry
#>   inf rec
#> S  -1   0
#> I   1  -1
#> R   0   1
```

The **stoichiometry matrix** shows the net change in each species for each transition. For infection, S decreases by 1 and I increases by 1; for recovery, I decreases by 1 and R increases by 1.

Visualising the Petri net

The `plot_petri()` function renders the Petri net as a bipartite graph via DiagrammeR. Species appear as blue circles and transitions as orange boxes:

```
plot_petri(sir)
#> Google Chrome was not found. Try setting the `CHROMOTE_CHROME` environment variable to the executabl
```



Generating odin2 code

The `pn_to_odin()` function translates the Petri net into odin2 DSL code. The default type is `"ode"` (continuous deterministic):

```
sir_code <- pn_to_odin(sir, type = "ode")
cat(sir_code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_inf <- inf * S * I
#> rate_rec <- rec * I
```

```
#>
#> ## Derivatives
#> deriv(S) <- -rate_inf
#> deriv(I) <- rate_inf - rate_rec
#> deriv(R) <- rate_rec
```

This generates mass-action kinetics where each transition rate is the product of the rate constant and the concentrations of input species.

Comparison with hand-written odin2

Here is the equivalent model written directly in the odin2 DSL:

```
sir_handwritten <- "
## Parameters
inf <- parameter()
rec <- parameter()

## Initial conditions
S0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(I) <- I0
initial(R) <- R0

## Mass-action rates
rate_inf <- inf * S * I
rate_rec <- rec * I

## Derivatives
deriv(S) <- -rate_inf
deriv(I) <- rate_inf - rate_rec
deriv(R) <- rate_rec
"
```

The generated code is structurally identical to the hand-written version. The algebraic approach guarantees that the stoichiometry is internally consistent—every molecule consumed by a transition is accounted for in the outputs.

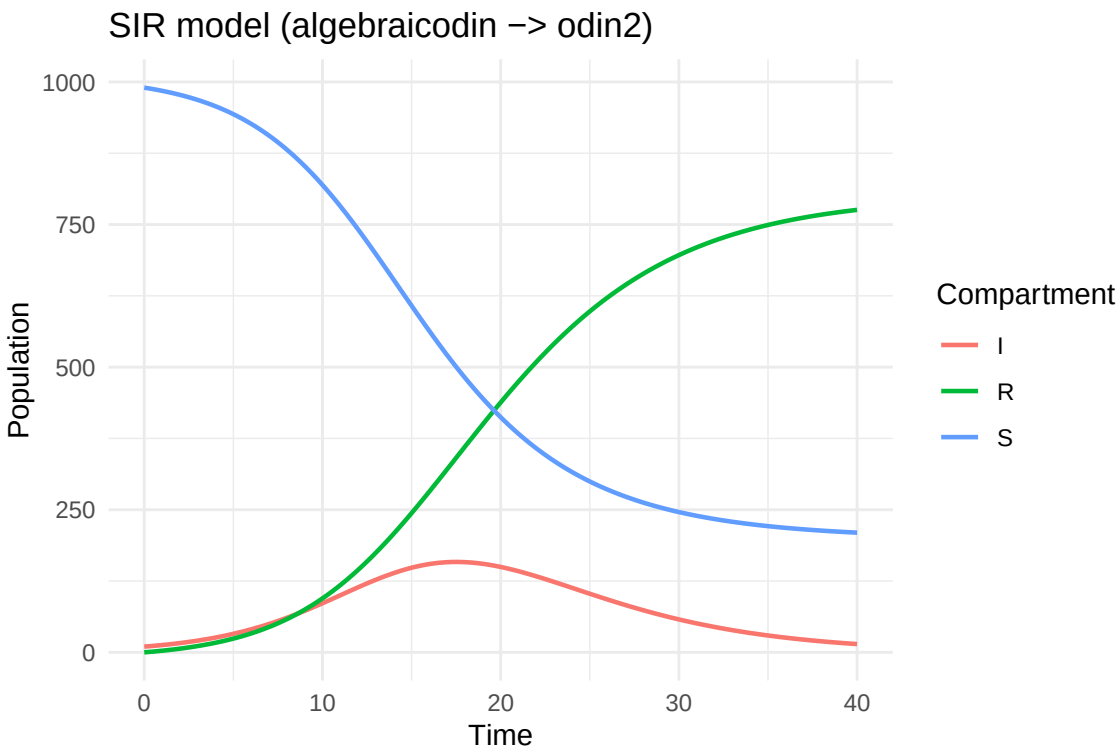
Running the model

We can compile and run the generated odin2 code using dust2:

```
gen <- odin2::odin(sir_code)
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system840d9d16
#> -- R CMD INSTALL ----------------------------------------------------------------------------------------
```

```
#>   -  installing *source* package ‘odin.system840d9d16’ ... (382ms)
#>    ** this is package ‘odin.system840d9d16’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:79:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                     ^
#>    dust.cpp:83:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       83 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mm1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system840d9d16) (496ms)
#>
#> i Loading odin.system840d9d16
pars <- list(inf = 0.0005, rec = 0.25, S0 = 990, I0 = 10, R0 = 0)
sys <- dust2::dust_system_create(gen, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys)
t <- seq(0, 40, by = 0.1)
y <- dust2::dust_system_simulate(sys, t)
```

```
df <- data.frame(
  time = rep(t, 3),
  value = c(y[1, ], y[2, ], y[3, ]),
  compartment = rep(c("S", "I", "R"), each = length(t))
)
library(ggplot2)
ggplot(df, aes(time, value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIR model (algebraicodin → odin2)",
       x = "Time", y = "Population", colour = "Compartment") +
  theme_minimal()
```



Using deSolve directly

For quick exploration without C++ compilation, the `vectorfield()` function generates a pure-R ODE function compatible with deSolve:

```
library(deSolve)
vf <- vectorfield(sir)
state <- c(S = 990, I = 10, R = 0)
parms <- c(inf = 0.0005, rec = 0.25)
out <- ode(state, t, vf, parms)
head(out)
#>      time        S        I         R
#> [1,]  0.0 990.0000 10.00000 0.0000000
#> [2,]  0.1 989.4990 10.24790 0.2530877
#> [3,]  0.2 988.9859 10.50167 0.5124432
#> [4,]  0.3 988.4603 10.76146 0.7782199
#> [5,]  0.4 987.9221 11.02737 1.0505676
#> [6,]  0.5 987.3708 11.29955 1.3296410
```

Summary

Approach	Pros	Cons
algebraicodin	Compositional; automatic stoichiometry; multi-target	Abstraction overhead for simple models
Hand-written odin2	Direct control; familiar syntax	Manual bookkeeping; error-prone for large models
deSolve	No compilation needed; immediate	Slower for large simulations; pure R

For small models, the approaches are essentially equivalent. The power of algebraicodin becomes clear when composing models from reusable parts (see `vignette("composition")`) or stratifying by age or risk groups (see `vignette("stratification")`).